Physical equilibrium

Learning outcomes

By the end of this section you should be able to describe the characteristics of a physical system at equilibrium.

You will begin this section by trying a simple activity that illustrates the concept of physical equilibrium. The procedure is as follows:

1. Fill two beakers half-full with water. Mark the water level using a marker.
2. Cover one of the beakers with plastic wrap and leave the other one open.
3. Place both beakers in a sunny place and leave them for a few days.

Figure 1 shows what the beakers might look like after a few days.

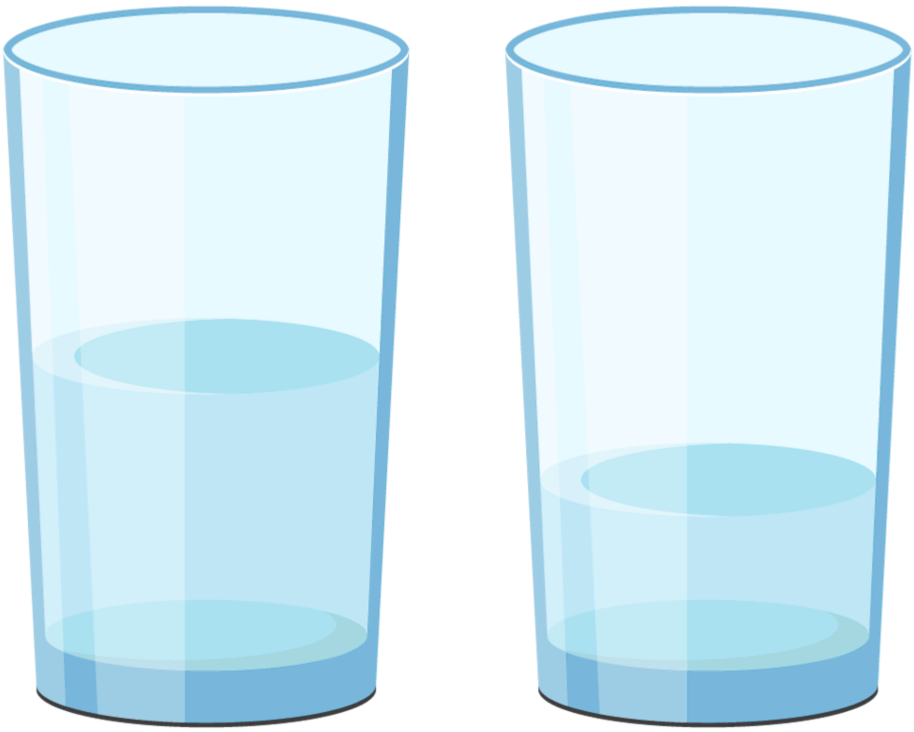


Figure 1. The beakers might look like this after a few days.

When you return to look at the beakers what do you notice? Has the water level changed in the beaker that was left open? What about in the beaker that was covered? How can you explain this? In the open beaker the water level has probably decreased quite considerably because the water evaporated. This can be represented as:

$$H_2O(l) \rightarrow H_2O(g)$$

The state symbol of the water has changed from (l) liquid to (g) gas. Also notice the use of a one-way arrow ($\rightarrow$) which suggests that only the forward process is occurring.

Now consider what happened in the beaker that was covered with the plastic wrap. You may have noticed that the water level has decreased slightly but nowhere near as much as the open beaker. Inside the beaker the water is evaporating but the reverse process, condensation, is also taking place. This can be represented as:

$$H_2O(l) \rightleftharpoons H_2O(g)$$

Although it might look similar to the previous equation, there is one very important difference, which is the use of a double arrow ($\rightleftharpoons$). This double arrow means that the forward process (evaporation) and the reverse process (condensation) are both taking place at the same rate. Once these two rates are equal, the system has reached what is known as physical equilibrium (plural equilibria). Physical equilibrium involves a physical change and occurs when the rate of the forward process is equal to the rate of the reverse process. The equilibrium is often referred to as being 'dynamic' because both forward and reverse processes are still occurring.

A closed system is required for a system to reach equilibrium. A closed system is one where only energy can be exchanged between the system and the surroundings and no matter can escape. This explains why the open beaker did not reach equilibrium but the sealed beaker did because the water was able to escape from the open beaker (an open system) but not from the sealed beaker (a closed system). **Figure 2** summarises the differences between an open system and a closed system.

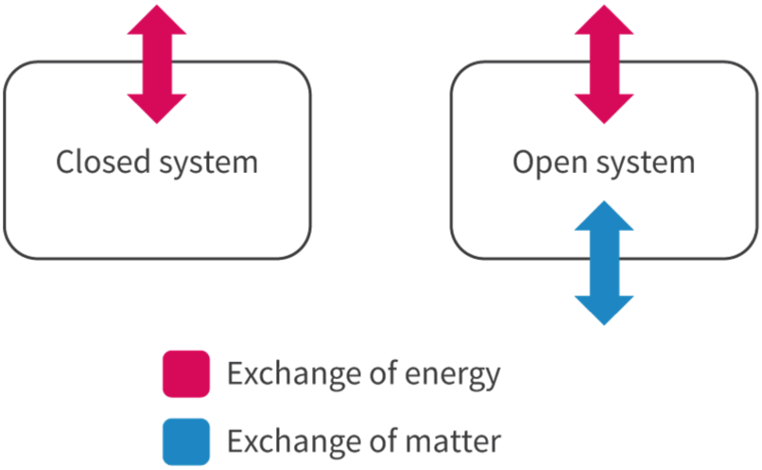


Figure 2. A closed system is required for a system to reach equilibrium.

More information for figure 2

The next example illustrates another important point about a system at equilibrium. Iodine, when heated, will sublime to form iodine vapour (**Video 1**). The iodine vapour will undergo deposition back to form solid iodine. This can be represented as:

$$I_2(s) \rightleftharpoons I_2(g)$$

Once the system reaches equilibrium, you will notice that the purple colour inside the flask remains constant. Colour is a macroscopic property meaning that it can be seen with the unaided eye. So for a system at equilibrium, there is no change in macroscopic properties.

Watch **Video 1** and note the point at which the system reaches equilibrium.

Chemistry experiment 47 - Sublimation of Iodine

Video 1. At equilibrium, the purple colour inside the beaker remains constant. This is a visible indicator that the system has reached equilibrium.

More information for video 1

Theory of Knowledge

Equilibrium in a chemical reaction can be seen as a competition between the formation of the products and formation of the reactants. Outside of chemistry, we also see opposing forces in many different contexts. In what situations does equilibrium exist outside the subject of science?

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

1. Write equations for the following physical equilibria.

   (a) The equilibrium between solid water and liquid water.
   (b) The equilibrium between liquid bromine and gaseous bromine.
   (c) The equilibrium between aqueous carbon dioxide and gaseous carbon dioxide.

2. Describe the type of system required for a physical equilibrium to be established.
3. Explain why clothes that are hung on a washing line on a hot day dry quickly.
4. Explain why wet clothes sealed in a plastic bag do not dry.
5. Bromine is a reddish-brown liquid which forms a similarly coloured vapour. Outline how you would be able to tell if an equilibrium has been established in a sealed flask continuing liquid bromine.

1. (a) $H_2O(s) \rightleftharpoons H_2O(l)$
   (b) $Br_2(l) \rightleftharpoons Br_2(g)$
   (c) $CO_2(aq) \rightleftharpoons CO_2(g)$
2. For a physical equilibrium to be established, a closed system is required. This is one in which only energy can be transferred between the system and the surrounding. Matter cannot escape which allows the equilibrium to be established.
3. Wet clothes dry quickly on a hot day because the rate of the evaporation is much greater than the rate of condensation. Effectively there is an open system which allows the matter (water) to escape which does not allow the equilibrium to be established.
4. Wet clothes sealed in a plastic bag do not dry because inside the bag, the rate of evaporation is equal to the rate of condensation. The sealed bag is a closed system which allows the equilibrium to be established.
5. Once the bromine starts to evaporate it would form a reddish-brown vapour. Once the reddish-brown colour (a macroscopic property) remains constant, you would know that an equilibrium has been established between the liquid bromine and the gaseous bromine.

R2.3 How far? The extent of chemical change

Chemical equilibrium

Learning outcomes

By the end of this section you should be able to describe the characteristics of a chemical system at equilibrium.

In section R2.3.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-equilibrium-id-43483/), a physical process described as occurring in a closed system will reach an equilibrium position in which the rate of the forward process is equal to the rate of the reverse process. In this section, you will learn about chemical equilibrium which involves a chemical change rather than a physical change, but what is the difference between the two? Use your knowledge of physical and chemical changes to complete the drag and drop activity in **Interactive 1**.

Physical changes	Chemical changes

Involve a physical change such as evaporation	Involve a change in the chemical composition of the matter
Involve the breaking of bonds between atoms	No change in the chemical composition of the matter
Involve a chemical reaction	Involve the overcoming of intermolecular forces between atoms or molecules
New products are formed	No new products are formed

Check

Interactive 1. Physical and Chemical Changes.

Keep these differences in mind as you study this section which focuses on chemical changes rather than physical changes.

A chemical system at equilibrium has much the same characteristics as a physical system at equilibrium but with one important difference - the system involves a chemical change rather than a physical change. One of the most important industrial chemical reactions is the Haber process (see section R2.3.0 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/)). In this reaction, hydrogen and nitrogen react together to form ammonia, NH_3.

The forward reaction is:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Kog

and the reverse reaction is:

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$

When nitrogen and hydrogen are mixed in a sealed reaction vessel, which is a closed system, the forward reaction proceeds as nitrogen and hydrogen react to produce ammonia. Once the concentration of ammonia starts to increase, the reverse reaction, in which ammonia decomposes to produce nitrogen and hydrogen, starts to take place. Eventually, the rate of the forward reaction will become equal to the rate of the reverse reaction. At this point, the system has reached equilibrium. The equilibrium is referred to as being dynamic because both the forward and reverse reactions are still taking place. This can be represented in graphical form as shown in **Figure 1**. Why do you think the reverse reaction starts from zero and then increases?

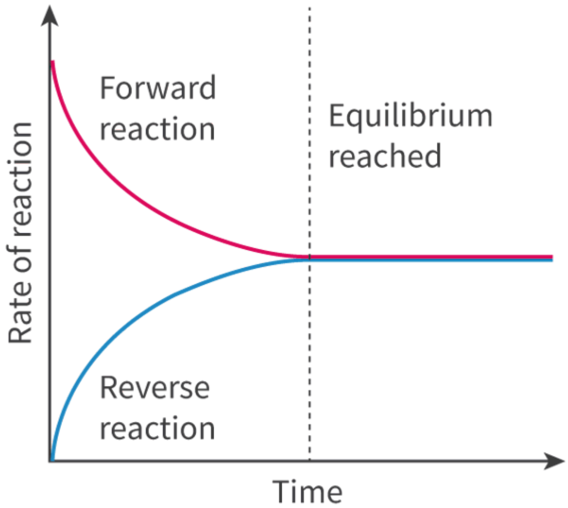


Figure 1. A chemical reaction reaches equilibrium when the rate of the forward reaction is equal to the rate of the reverse reaction.

More information for figure 1

Initially, the concentration of the product (ammonia) is zero, so there aren't any molecules to participate in the reverse reaction, which means the rate is also zero. As the concentration of the ammonia increases, the rate of the reverse reaction also increases.

The next characteristic of a chemical reaction at equilibrium involves the concentrations of reactants and products. The rate of reaction is defined as the change in concentration per unit time section R2.2.1

Kog

(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-rate-of-reaction-id-43481/). How could the rate of reaction for the Haber process be followed in which all species are gases?

The rate of reaction could be monitored by the change in concentration of the reactants or of the products.

But what would this look like when both the forward and reverse reactions are occurring?

Figure 2 shows a graph of the change in the concentrations of nitrogen, hydrogen and ammonia as the forward reaction proceeds. The concentrations of nitrogen and hydrogen decrease as they react together and produce ammonia and the concentration of ammonia starts from zero and increases as initially the reaction vessel only contains nitrogen and hydrogen. What do you notice about the concentrations of the reactants and products once the reaction reaches equilibrium?

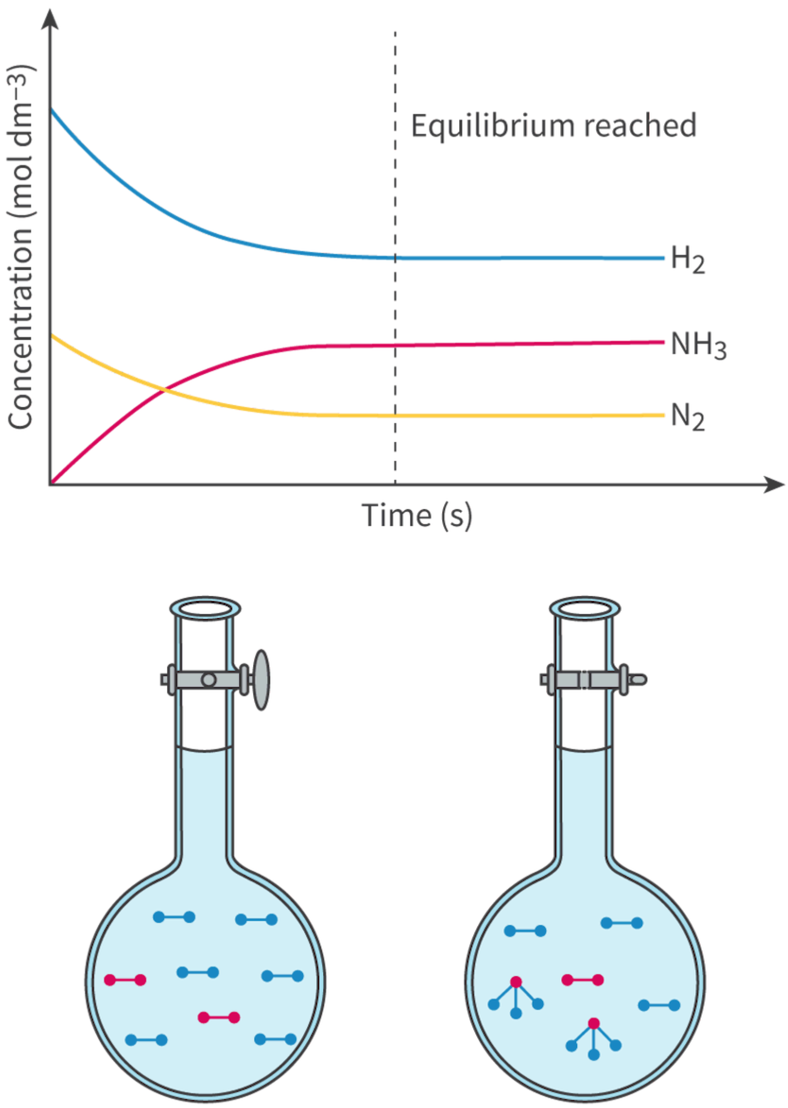


Figure 2. Once a reaction reaches equilibrium the concentrations of reactants and products remain constant.

More information for figure 2

You may have noticed that once the reaction reaches equilibrium, the concentrations of the reactants and products do not change, in other words they remain constant. This is an important point about a reaction at equilibrium: the rates of the forward and reverse reactions are equal but the concentrations of reactants and products are not. It is also important to note that at equilibrium, the concentration of reactants and products are rarely equal. It is much more common to have higher concentrations of reactants or products when a reaction reaches equilibrium.

A chemical reaction can reach equilibrium from either direction starting with the reactants or with the products. **Figure 3** and **Figure 4** show this for the reaction between sulfur dioxide and oxygen to produce sulfur trioxide and also for the reverse reaction in which sulfur trioxide reacts to form sulfur dioxide and oxygen.

The equation for the forward reaction is:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

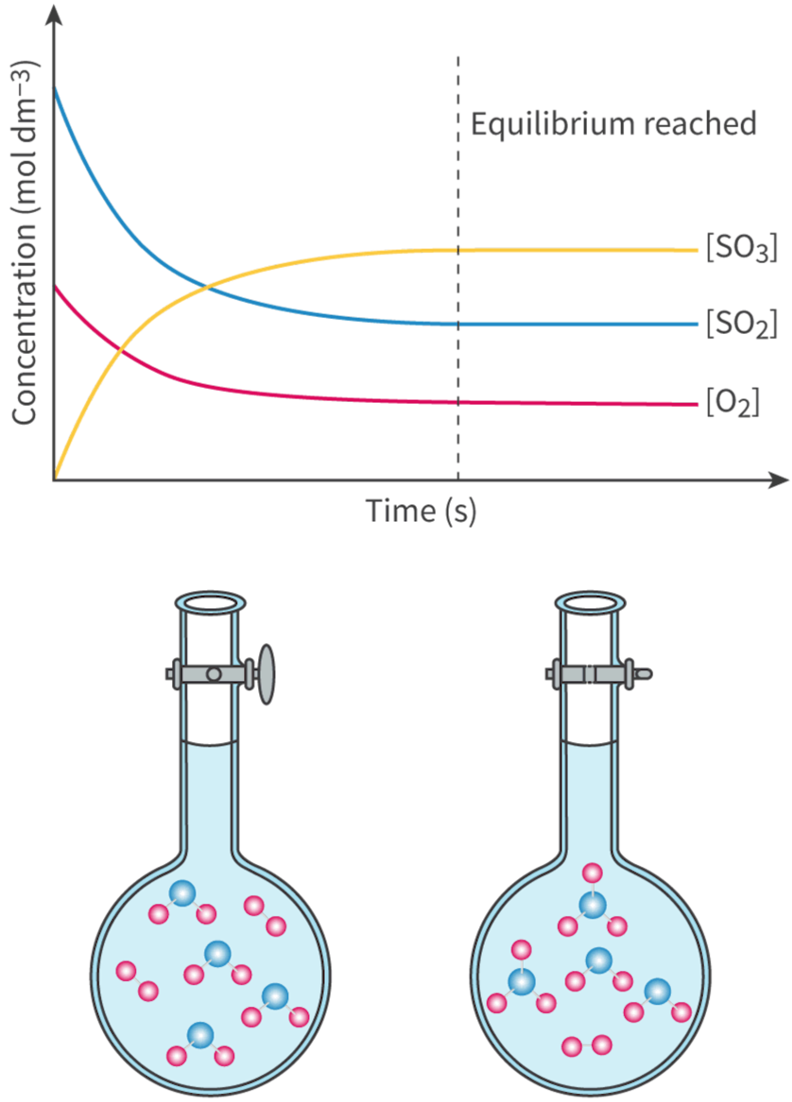
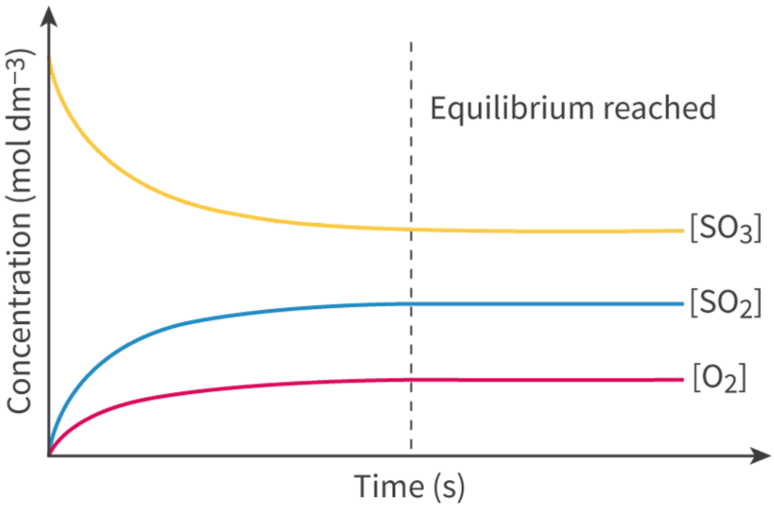


Figure 3. The equilibrium position can be reached from the forward reaction or the reverse reaction.

More information for figure 3

And for the reverse reaction:

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$$



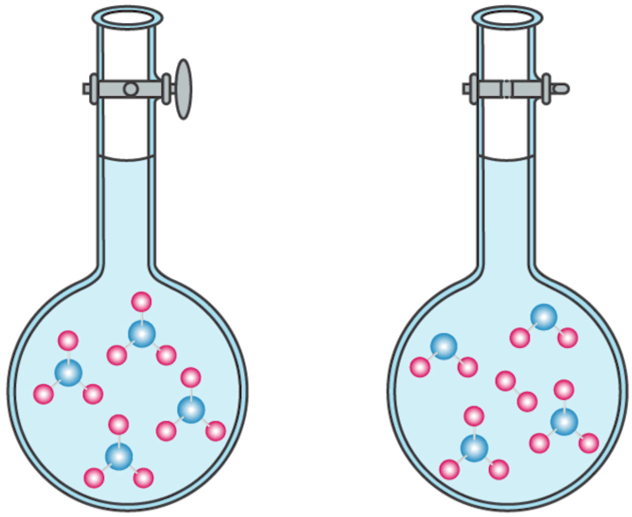


Figure 4. The equilibrium position can be reached from the forward reaction or the reverse reaction.

More information for figure 4

From the graphs, you can see that from whichever side the reaction starts (reactants or products), the equilibrium position is reached when the concentration of reactants and products remain constant.

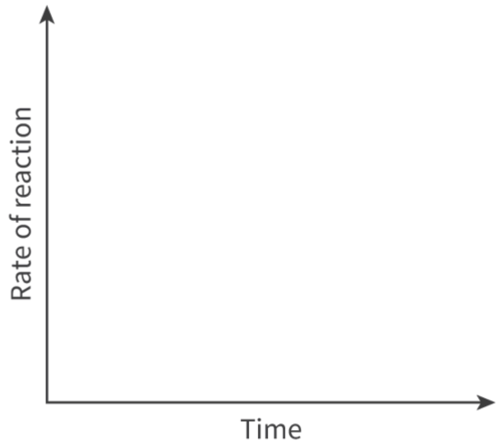
Study skills

You might have noticed that all the examples of equilibrium covered so far involve reaction where the reactants and products are in the same state. These are known as homogeneous reactions or examples of homogeneous equilibria. There are also equilibrium systems where the reactants and products are in different states, known as heterogeneous equilibrium, such as solid ice cubes and liquid water.

Activity

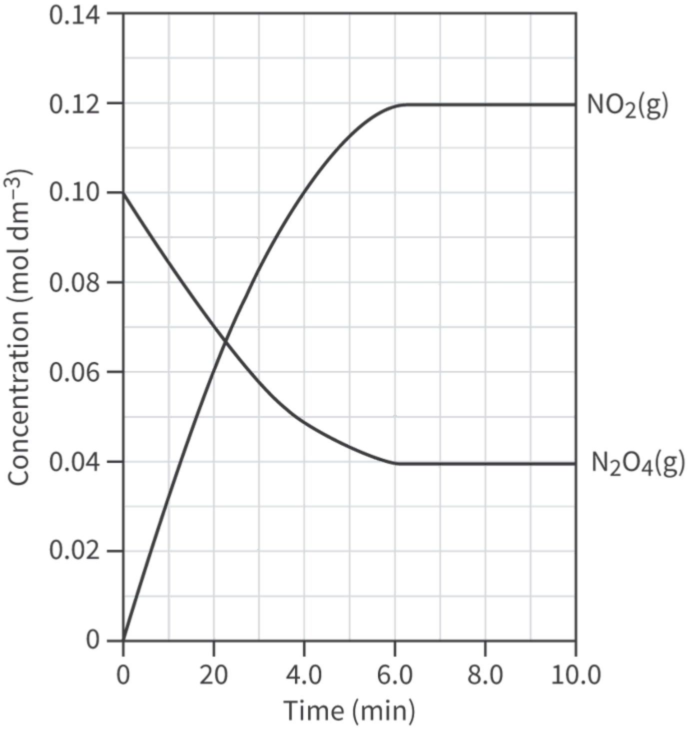
- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

1. When a chemical reaction is at equilibrium, state how the rate of the forward reaction compares to the rate of the reverse reaction.
2. Outline the difference between using a one-way arrow (→) and a double arrow (⇌) for a chemical reaction.
3. Sketch a graph on the axis below to represent the forward and reverse reactions for a reversible reaction. Label the point at which the reaction reaches equilibrium.



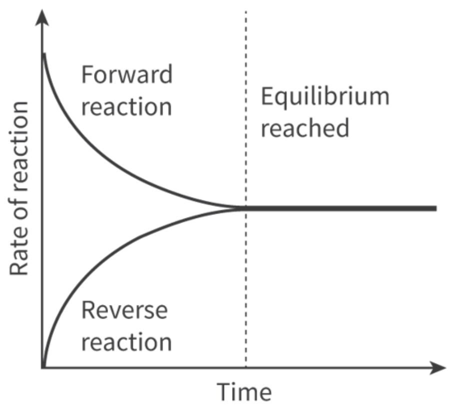
More information

4. The graph below shows how the concentrations of NO_2(g) and N_2O_4(g) change throughout the course of a reaction. Determine the time at which the reaction reaches equilibrium giving a reason for your answer.



More information

1. For a reaction at equilibrium, the rate of the forward reaction is equal to the rate of the reverse reaction.
2. A one-way arrow implies that the reaction proceeds in the forward direction only. A double arrow implies that the reaction is reversible meaning that both the forward and reverse reactions occur.
3. The graph should look like the one shown below. The position of equilibrium is reached once the rate of the forward reaction is equal to the rate of the reverse reaction.



4. The reaction reaches equilibrium at six minutes. The concentrations of reactants and products remain constant at equilibrium.